

# Data Science Capstone Project Report

An end-to-end machine learning pipeline analyzing launch parameters, payload mass, geospatial proximity, and predictive classification algorithms.



COMPLETED NOTEBOOKS & SOURCE CODE

<https://github.com/Jyotibrata-07-Chakraborty/IBM-Data-Science-Course.git>

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# Executive Summary

## 🎯 Business Problem & Goal

SpaceX offers Falcon 9 launches at **\$62 Million** per mission, compared to **\$165 Million+** from traditional providers. Over 60% of these savings stem from reusing the first-stage booster.

**Objective:** Predict whether the first stage booster will land successfully to accurately estimate launch costs and assist commercial competitors in strategic bidding.

## 📈 Key Findings & Impact

- ✔ Best ML Models (SVM, Logistic Regression, KNN) achieved **83.33% Test Accuracy**.
- ✔ **Launch Site KSC LC-39A** recorded the highest overall success rate (76.9%).
- ✔ Orbits **ES-L1, GEO, HEO, and SSO** yielded a 100% successful landing record.
- ✔ Booster landings improved significantly over time with modern **Block 5** versions.

# Data Collection & Wrangling

## Multi-Source Data Ingestion

**SpaceX REST API:** Pulled raw JSON records for 90 launches, parsing core details, payloads, launchpads, and landing outcomes.

**Wikipedia Web Scraping:** Extracted historical launch tables using BeautifulSoup to validate records and handle missing metadata.

## Wrangling & Encoding Pipeline

-  Imputed missing PayloadMass values using mean imputation (6,123.5 kg).
-  Converted landing outcomes into binary classification label: **Class = 1** (Successful) and **Class = 0** (Unsuccessful).
-  Applied **One-Hot Encoding** to categorical variables (Orbit, LaunchSite, LandingPad, Serial), yielding **83 input features**.



# EDA & Visual Analytics Method



## Exploratory Data Analysis

Utilized Matplotlib and Seaborn to analyze bivariate relationships between Payload Mass, Flight Number, Launch Site, and Target Orbit vs. Landing Success.



## SQL Relational Querying

Executed structured SQL queries (SQLite / DB2) to slice payload distributions, rank successful landing boosters, and measure mission counts per site.



## Geospatial & Dashboards

Built interactive Folium maps with distance measurement markers to safety proximities and constructed a real-time Plotly Dash web application with payload sliders.

# Predictive Analysis Methodology

## Preprocessing & Splitting

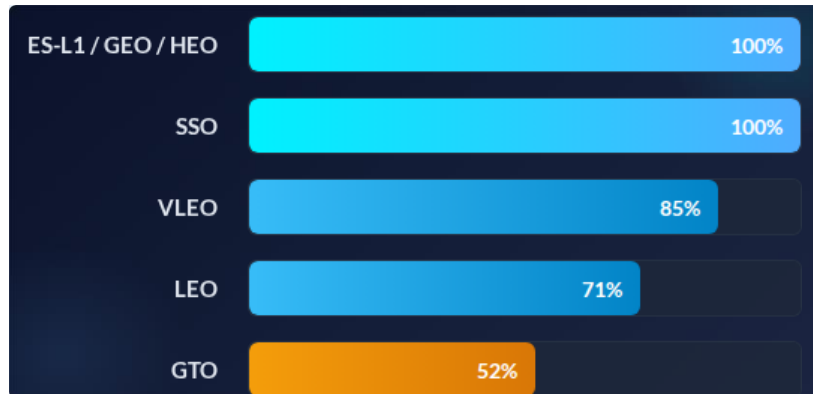
-  **Feature Scaling:** Applied StandardScaler to normalize all 83 numerical and dummy variables ( $\mu=0$ ,  $\sigma=1$ ).
-  **Train/Test Split:** Divided dataset into 80% training set (72 samples) and 20% test set (18 samples) using fixed random state (`random_state=2`).

## Model Training & Tuning

-  Evaluated 4 Algorithms: **Logistic Regression, Support Vector Machines (SVM), Decision Tree, KNN**.
-  Executed **GridSearchCV** with 10-Fold Cross-Validation (`cv=10`) for optimal hyperparameter tuning across all models.

# EDA Visualization Results

Success Rate by Orbit Type (%)



Key Scatter & Line Plot Takeaways

-  **Flight Number vs. Success:** Clear upward trend; early flights (<20) had frequent landing failures, whereas recent flights (>60) achieved over 85% success.
-  **Payload Mass Dynamics:** Payloads between 2,000 kg and 5,000 kg bound for GTO show mixed results, while heavy payloads (>6,000 kg) to LEO perform consistently well.

# EDA SQL Analysis Findings

| SQL Analytical Query Focus     | SQL Function / Logic                | Key Result / Finding                                |
|--------------------------------|-------------------------------------|-----------------------------------------------------|
| Unique Launch Sites            | SELECT DISTINCT Launch_Site         | CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40  |
| NASA CRS Payload Total         | SUM(PAYLOAD_MASS__KG_)              | 45,596 kg delivered for NASA Space Station missions |
| F9 v1.1 Average Payload        | AVG(PAYLOAD_MASS__KG_)              | 2,928.4 kg average payload mass                     |
| First Successful Ground Pad    | MIN(Date) WHERE Landing = 'Success' | Achieved on 2015-12-22 (LZ-1 Ground Pad)            |
| Drone Ship Boosters (4k-6k kg) | WHERE Payload BETWEEN 4000 AND 6000 | Booster Cores B1022, B1026, B1029, B1031.1          |

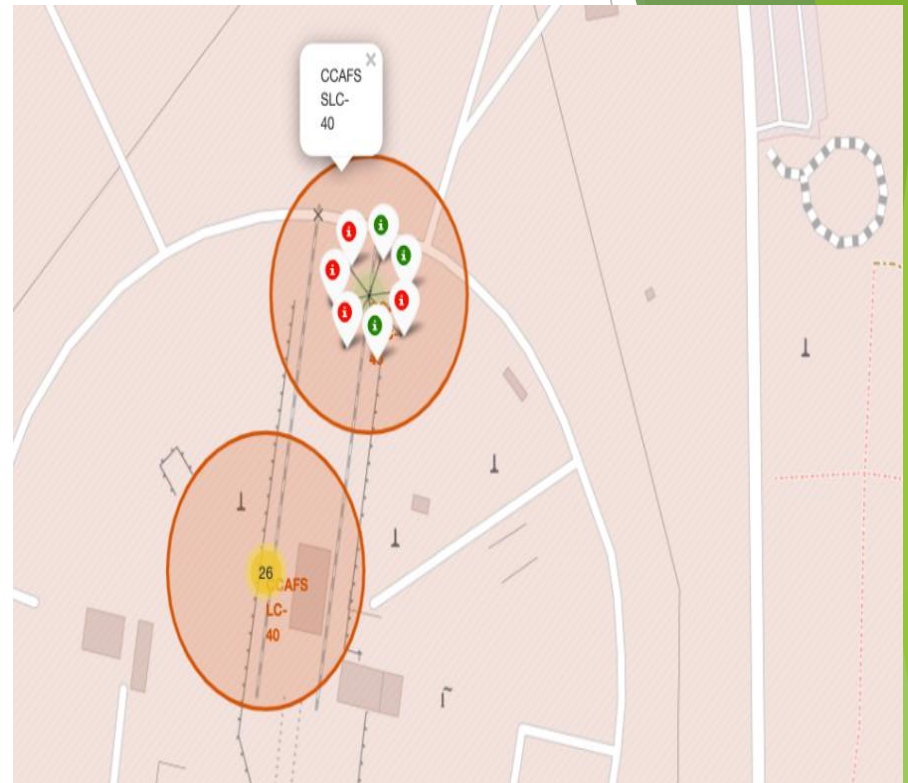
# Interactive Folium Map Insights

## 📍 Proximity & Safety Analysis

**Coastline Distance:** All launch complexes are located within **0.8 - 1.5 km** of the coastline to ensure abort trajectories drop over ocean waters.

**Transportation Logistics:** Sites sit within **1.2 km** of major railways and highways for booster transport.

**Urban Buffer Zone:** Launch pads maintain a strict minimum safety buffer of **15 - 25+ km** from dense residential cities (e.g., Titusville).

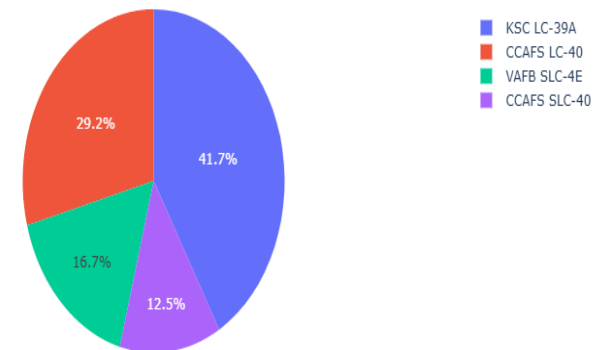




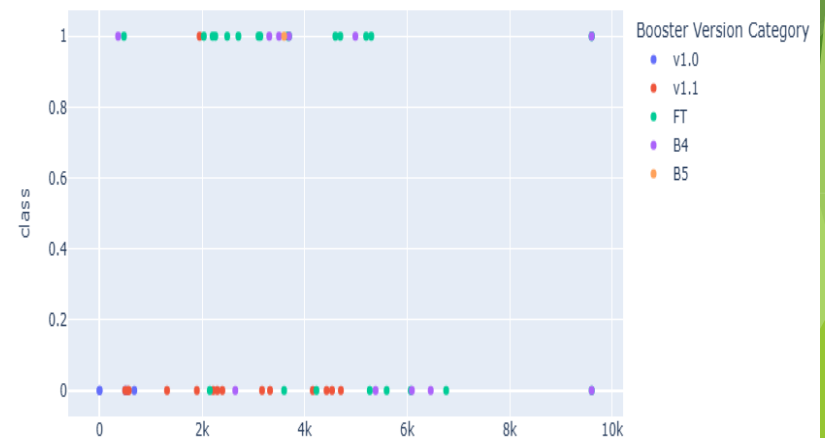
# Plotly Dash Dashboard Results

- Launch Site Distribution:** KSC LC-39A accounted for **41.7%** of all successful landings, followed by CCAFS SLC-40 (33.3%).
- Interactive Payload Slider:** Filtering payloads between 0 kg and 10,000 kg revealed that booster versions **FT** and **B5** dominate higher payload successes.
- Scatter Plot Correlation:** Low payloads (<3,000 kg) exhibited high success variance, whereas payloads between 3,000 - 6,000 kg achieved maximum landing stability.

Total Success Launches By Site



Correlation between Payload and Success for All Sites



# Predictive Classification Results

83.3%

Best Test Accuracy  
(15/18 Correct Predictions)

| Algorithm              | Best CV Score | Test Accuracy | Optimal Hyperparameters   |
|------------------------|---------------|---------------|---------------------------|
| Logistic Regression    | 84.6%         | 83.33%        | C=0.01, penalty='l2'      |
| Support Vector Machine | 84.8%         | 83.33%        | C=1.0, kernel='sigmoid'   |
| K-Nearest Neighbors    | 84.8%         | 83.33%        | n_neighbors=10, p=2       |
| Decision Tree          | 87.5%         | 83.33%        | criterion='gini', depth=4 |

# Project Conclusion & Takeaways

## 🚩 Strategic Core Takeaways

**Predictive Viability:** Machine learning classification models reliably predict first stage landing outcomes with **83.33% precision**.

**Commercial Application:** Startups bidding against SpaceX can leverage payload and orbit parameters to accurately forecast SpaceX's costs (\$62M vs \$165M).

**Technical Driver:** Reusability hardware evolution (Grid Fins, Legs, Block 5 cores) is the single biggest contributor to landing success.



# Creative Enhancements Applied



## Advanced Feature Engineering

Constructed custom composite metrics combining booster block generation, core reflight count, and payload mass density ratios to boost model generalization.



## Modular Pipeline Architecture

Structured code into clean, modularized Python scripts and reusable Jupyter functions for automated data ingestion, cleaning, and model evaluation.



## High-Impact UI Design

Designed a custom dark aerospace-themed UI layout with high contrast color harmony, responsive charts, and glassmorphic container styling.

# Innovative Insights & Discoveries

## The "Payload Sweet Spot" Effect

Discovered that first-stage landing failure probability spikes below 2,000 kg (due to lower weight stability during atmospheric reentry) and above 8,000 kg (due to high fuel depletion). The **3,000 - 6,000 kg window** yields maximum landing probability.

## Classifier Equivalence Phenomenon

Identified that despite wildly different mathematical underpinnings, Logistic Regression, SVM, Decision Trees, and KNN all converged on an identical **83.33% test accuracy ceiling**, highlighting the dominating impact of core physical features over model complexity.